Ninjacart Vegetable Classifier

AI-Driven Fresh Produce Supply Chain Automation

This project implements a Deep Learning based multiclass image classifier designed for **Ninjacart**, India's largest fresh produce supply chain company. The goal is to automate the vegetable sorting process to meet a strict 12-hour farm-to-store delivery constraint.

TensorFlow/Keras · Transfer Learning · Computer Vision · Supply Chain AI

Project Overview

The system distinguishes between three primary vegetable types (Onions, Potatoes, Tomatoes) and handles non-vegetable market scenes as "noise". This ensures only high-quality produce enters the automated packaging and delivery pipeline.

Key Features

- **Noise Detection:** Automatically filters out irrelevant images (market scenes) to prevent system errors.
- **Fine-Grained Classification:** High-precision differentiation between visually similar items like onions and potatoes.
- **Model Benchmarking:** Comparative analysis of Custom CNN, VGG16, VGG19, and ResNet101.
- **Scalable Architecture:** Designed for rapid inference to support real-time conveyor belt sorting.

Performance Metrics

Model Architecture	Test Accuracy	Notes
VGG19	80.63%	Most stable; excellent generalization.
VGG16	79.49%	Fastest convergence.
Custom CNN	78.06%	Good performance but more volatile.
ResNet101	38.75%	Too complex for current dataset size.

Technical Implementation

Prerequisites

```
import tensorflow as tf
import numpy as np
import cv2
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

How to Run

1. **Dataset Preparation:** Place training and testing images in `ninjacart_data/train` and `ninjacart_data/test`.
2. **Data Pipeline:** Use the `DataPipeline` class for automated rescaling (1./255) and augmentation.
3. **Training:** Execute the `run_model` function with your chosen architecture (e.g., `VGG19`).
4. **Evaluation:** Generate confusion matrices and accuracy/loss curves using the `ModelEvaluator`.

Business Impact

- **12-Hour Promise:** Eliminates manual sorting bottlenecks to ensure produce reaches retailers within hours of harvest.

- **Quality Standardization:** Removes human subjectivity and fatigue from the grading process.
- **Traceability:** Creates a digital audit trail for quality control and inventory management.
